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Barrier Integrity and Immunity: Exploring the Cutaneous Front Line in Health and Disease

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Keywords

tight junction, corneoptosis, stratum corneum, pH, hair follicle, skin microbiota

Abstract

Immune responses are influenced by not only immune cells but also the tissue microenvironment where these cells reside. Recent advancements in understanding the underlying molecular mechanisms and structures of the epidermal tight junctions (TJs) and stratum corneum (SC) have significantly enhanced our knowledge of skin barrier functions. TJs, located in the granular layer of the epidermis, are crucial boundary elements in the differentiation process, particularly in the transition from living cells to dead cells. The SC forms from dead keratinocytes via corneoptosis and features three distinct pH zones critical for barrier function and homeostasis. Additionally, the SC–skin microbiota interactions are crucial for modulating immune responses and protecting against pathogens. In this review, we explore how these components contribute both to healthy and disease states. By targeting the skin barrier in therapeutic strategies, we can enhance its integrity, modulate immune responses, and ultimately improve outcomes for patients with inflammatory skin conditions.

Atopic dermatitis

(AD): a chronic and relapsing heterogeneous inflammatory skin disorder that is characterized by skin rashes and itch; it is primarily driven by Th2-derived cytokines

Psoriasis: a chronic inflammatory skin disorder that is defined by raised plaques and scales; it is driven by Th17-derived cytokines

Filaggrin: a protein that aggregates keratin and degrades into a natural moisturizing factor; its precursor, profilaggrin, aggregates to form keratohyalin granules

1. INTRODUCTION

Inflammatory skin diseases encompass a diverse array of conditions characterized by inflammation and compromised integrity within the skin's complex architecture. These disorders have a significant impact on the quality of life of affected individuals, who present with different clinical symptoms of varying severity and with progression across diseases such as atopic dermatitis (AD), psoriasis, and other dermatoses. Among these diseases, AD is especially prevalent and affects individuals of all ages and ethnicities worldwide. It is marked by dry, sensitive skin and recurrent eczematous lesions accompanied by intense itching (1–3). This condition manifests in infancy or childhood in approximately 80% of cases, with the remaining cases developing in adulthood (1, 4, 5).

The identification of filaggrin gene (*FLG*) mutations as significant genetic risk factors for AD has been pivotal in revolutionizing our understanding of the condition (6, 7). This discovery not only underscored the crucial role of filaggrin in maintaining the structural integrity of the skin barrier but also shed light on aspects of skin barrier biology that previously were not well understood. Filaggrin is essential for skin hydration and protection against external irritants. Loss-of-function mutations in *FLG* can compromise skin barrier function, leading to increased permeability and reduced moisture retention. Such an impaired skin barrier allows allergens to penetrate the skin more easily (8), triggering immune responses and the inflammatory characteristics observed in AD (**Table 1**) (9). The discovery of filaggrin mutations highlighted the importance of epidermal structure and skin barrier integrity, illustrating how a breach in this primary physical barrier can provoke immune reactions, even in the presence of an otherwise intact immune system.

The last two decades have seen significant advances in our understanding of the skin barrier, particularly of the roles of tight junctions (TJs) and the stratum corneum (SC). These components are crucial for maintaining overall skin health and integrity. The integrity of the skin barrier is paramount in regulating immune responses within the skin microenvironment. Dysfunction in the skin barrier is a primary driver of inflammation in conditions like AD, highlighting the critical interplay between the skin's structural components and immune regulation. Thus, by enhancing our knowledge of these key components, researchers can develop targeted interventions to strengthen the skin barrier, modulate immune responses, and alleviate symptoms of skin diseases.

Continued research into these areas is crucial for advancing personalized treatment approaches. Understanding and harnessing the recent advancements in skin barrier integrity are essential for improving outcomes for individuals with inflammatory skin diseases, such as AD. As our comprehension of these mechanisms expands, new avenues will open for therapeutic strategies aimed at not just treating the symptoms but also fundamentally altering the course of these diseases through improved skin barrier function.

2. THE STRUCTURE OF THE SKIN

The skin, a sophisticated and multilayered organ, serves as the body's primary defense against external elements and plays a vital role in protection, sensation, and temperature regulation (10). It consists of three main layers: the epidermis, dermis, and hypodermis (**Figure 1a,b**). In terrestrial vertebrates, the body surface is sealed by the stratified squamous epithelium of the skin, termed the epidermis, which is equipped with two sets of protective barriers: the SC and TJs (11) (**Figure 1a,b**). The dermis, which lies just beneath the epidermis, provides the skin with flexibility and strength through a dense network of collagen fibers and elastin fibers. This layer contains blood vessels, sensory nerve endings, hair follicles, and sweat glands, all of which are nestled within a supportive matrix that maintains the structural integrity of the skin. Below the dermis is the hypodermis or subcutaneous layer, consisting of loose connective tissue and fat pads that act as insulation and cushioning, absorbing shocks and preserving body heat.

Table 1 Skin barrier-related genes associated with AD, AD-like dermatitis, or Th17-driven dermatitis

Gene	Protein	Function(s)	Related human disease	Human skin phenotype		KO mouse phenotype	References
				AD-like dermatitis	Th17-driven dermatitis		
Filaggrin system							
<i>FLG</i>	Filaggrin	Serves as major constituent of KHGs Bundles keratin filaments Produces NMFs	Ichthyosis vulgaris, AD	Yes	ND	Decreased NMFs and dry scaly skin Increased antigen penetration of the SC Premature shedding of SC layers and increased desquamation under mechanical stress	6–9
Formation of CCE							
<i>TGM1</i>	Transglutaminase 1	Cross-links CCE proteins	ARCI1 (LI)	ND	Yes	Neonatal lethal Abnormal CCE and intracellular lipid lamellae	73, 82, 93–97
Formation of lipid lamellae and/or CLE							
<i>ALOX12B</i>	12R-lipoxygenase	Processes ceramide	ARCI2 (CIE)	ND	Yes	Neonatal lethal Abnormal SG and fragile CLE Aberrant SC lipid composition	44, 73, 93–97
<i>ALOXE3</i>	Epidermal lipoxygenase 3	Processes ceramide	ARCI3 (CIE)	ND	Yes	ND	44, 73, 93–97
<i>TMEM79</i>	Tmem79/mattirin	Secretes lamellar bodies	AD	Yes	Yes	Spontaneous dermatitis and increased IgE levels Atopy and increased cutaneous sensitization Impaired SC formation and impaired lamellar body Impaired lamellar body protein secretion	60, 78, 79

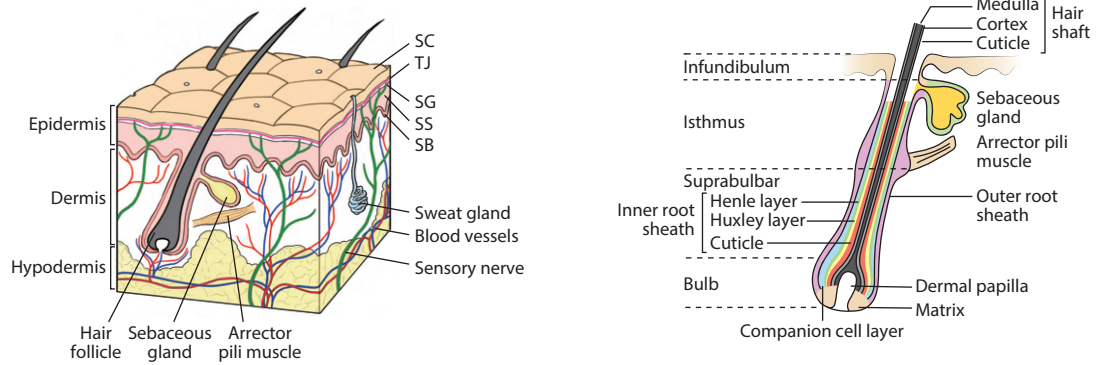
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Table 1 (Continued)

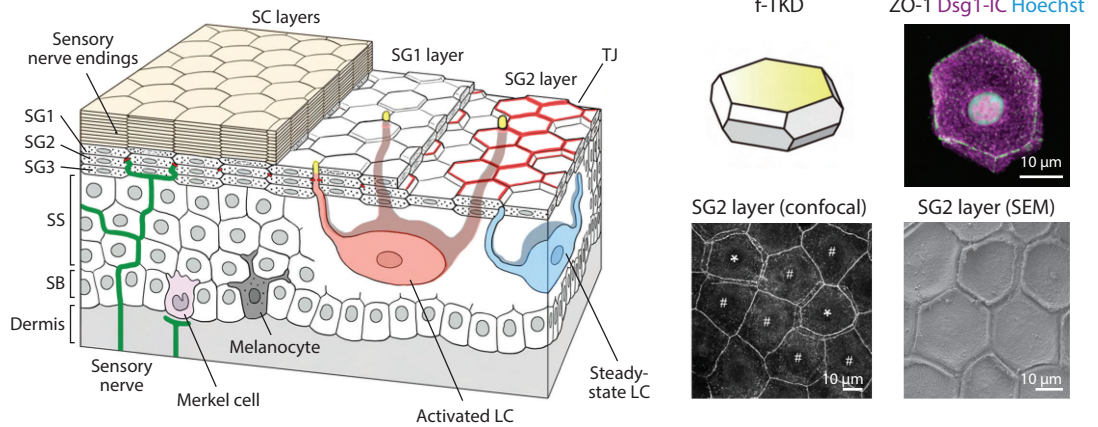
Gene	Protein	Function(s)	Related human disease	Human skin phenotype		KO mouse phenotype	References
				AD-like dermatitis	Th17-driven dermatitis		
Corneocyte desquamation							
<i>SPINK5</i>	Serine peptidase inhibitor Kazal type 5	Inhibits kallikrein-related peptidases	Netherton syndrome	Yes	Yes	Neonatal lethal; fragile skin with dermatitis	11, 62, 73, 93–97
Corneodesmosome							
<i>CSDN</i>	Corneodesmosin	Supports the corneodesmosome	Peeling skin syndrome 1	Yes	ND	Neonatal lethal due to tearing Desmosomal break between the SG and SC Hyperproliferation of keratinocytes and development of thick SC	11, 47, 62
<i>DSG1</i>	Desmoglein 1	Serves as a cadherin family member	SAM syndrome	Yes	ND	ND	61, 62
Tight junction							
<i>CLDN1</i>	Claudin-1	Serves as an integral transmembrane protein of TJJs	NISCH syndrome	Yes	ND	Neonatal lethal (complete KO) Spontaneous dermatitis (epidermal cKO) Abnormal SC and SC lipid composition (both)	11, 20, 23, 24

Abbreviations: AD, atopic dermatitis; ARCI, autosomal recessive congenital ichthyosis; CCE, cornified cell envelope; CIE, congenital ichthyosiform erythroderma; CLE, cornified lipid envelope; KHG, keratohyalin granule; KO, knockout; LI, lamellar ichthyosis; ND, not determined; NISCH, neonatal ichthyosis sclerosing cholangitis; NMFs, natural moisturizing factors; SAM, severe dermatitis, multiple allergies, and metabolic wasting; SC, stratum corneum; SG, stratum granulosum; TJ, tight junction.

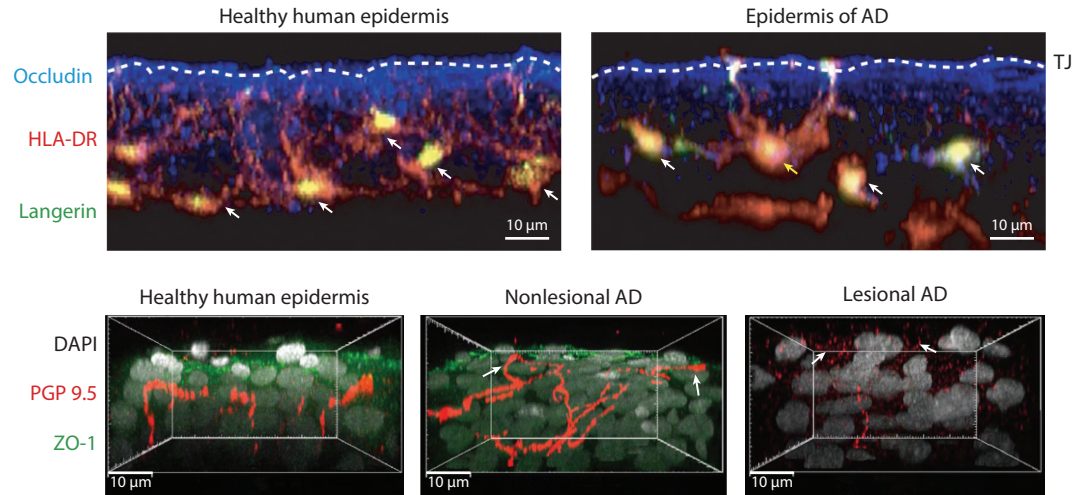
a Structure of skin and hair follicle



b Healthy skin



c Comparison of healthy skin and AD skin



(Caption appears on following page)

Figure 1 (Figure appears on preceding page)

Skin barrier components and their structures in health and disease. (a) Overview of the skin (*left*). The skin consists of three main layers: the epidermis, dermis, and hypodermis. The epidermis is equipped with two sets of protective barriers: the SC and TJ. The dermis contains blood vessels, the sensory nerve, hair follicles, and sweat glands. The hypodermis consists of loose connective tissue and fat pads. An overview of the hair follicle is shown on the right. (b) Illustration of skin barrier components (the SC, the TJ, and the LC network) of healthy human skin (*left*). In healthy human skin, corneocytes in the SC and keratinocytes in the SG are shaped like an f-TKD, and TJs form on the edge of the f-TKD of SG2 cells (*top middle*). Normal LCs remain below the TJ and project their dendrites (*blue*), but activated LCs extend their dendrites beyond the TJ to capture external antigens that penetrate the SC (*red*). Furthermore, epidermal nerve endings extend and retract while remaining under the TJ barrier. (*Top right*) 3D-rendered image of isolated SG2 cells visualized by cytoplasmic staining for Dsg1-IC, with TJ component ZO-1 at the edges. (*Bottom middle*) X-Y plane confocal microscopic image of a ZO-1-positive honeycomb in the mouse ear epidermis displaying double-edged (*asterisks*) and single-edged (*number signs*) polygons. (*Bottom right*) X-Y plane SEM image of the surface of the SG2 layer showing polyhedral-shaped keratinocytes. Top middle, top right, and bottom middle sections of panel *b* adapted from Reference 13 (CC BY 4.0). (c) Confocal microscopic images of TJs and LCs of healthy human skin (*top left*) and AD skin (*top right*). The images show staining for occludin, HLA-DR, and langerin. The yellow arrow indicates activated LCs that extend their dendrites beyond the TJ, and the white arrows show steady-state LCs that do not extend their dendrites beyond the TJ. Panel *c*, top, adapted from Reference 29 with permission from Elsevier. (*Bottom*) 3D confocal microscopic images of epidermal nerve endings and TJs in healthy human skin, nonlesional AD skin (*bottom middle*) and lesional AD skin (*bottom right*); none of the nerve fibers penetrated the TJs in the healthy human skin (*bottom left*). Furthermore, nonlesional AD skin and lesional AD skin exhibited low and very low ZO-1 expression levels, respectively. The white arrows indicate epidermal nerve endings extended outside the TJs. Panel *c*, bottom, adapted from Reference 39 (CC BY 4.0). Abbreviations: AD, atopic dermatitis; DAPI, 4',6-diamidino-2-phenylindole; Dsg1-IC, intracellular portion of desmoglein 1; f-TKD, flattened Kelvin's tetrakaidecahedron; LC, Langerhans cell; PGP, protein gene product; SB, stratum basale; SC, stratum corneum; SEM, scanning electron microscopy; SG, stratum granulosum; SS, stratum spinosum; TJ, tight junction; ZO, zonula occludens.

At the surface, the epidermis forms the skin's primary protective barrier and is predominantly composed of keratinocytes, which are organized into four distinct layers (**Figure 1b**). The deepest layer, the stratum basale, hosts basal cells, which divide continuously to generate new keratinocytes; melanocytes, which produce melanin pigments; and Merkel cells, which are involved in touch sensation. Located upward in the skin, keratinocytes in the stratum spinosum develop spiny extensions owing to desmosomal connections and thereby enhance the structural integrity of the skin. This layer also accommodates Langerhans cells (LCs), which are integral to the body's immune system and defense mechanisms.

In the stratum granulosum (SG), which is composed of three layers (from outer to inner, SG1, SG2, and SG3), keratinocytes form a TJ barrier at the SG2 layer and release lipids to fill the extracellular spaces around the SG1 cells, establishing a waterproof barrier that prevents fluid loss and protects against pathogens. Furthermore, SG1 cells undergo a type of programmed cell death termed corneoptosis (12) and convert into nonviable enucleated keratinocytes (corneocytes) to form the SC, the outermost layer of the epidermis. Finally, the SC consists of corneocytes shaped like a flattened Kelvin's tetrakaidecahedron (f-TKD) (13–15), forming a robust shield against environmental threats such as bacteria, viruses, and contaminants (**Figure 1b**).

Hair follicles, which are extensions of the epidermis and penetrate the dermis (**Figure 1a**), produce hair and are associated with sebaceous glands that secrete oils for skin and hair conditioning. Sweat glands—including eccrine glands, which are distributed across the body surface, and apocrine glands, which are located in hair follicle-rich regions such as the armpits and groin—regulate body temperature and eliminate waste products in response to emotional and hormonal cues.

Collectively, these intricate layers and structures ensure effective skin functions and safeguard the body while dynamically engaging with the external environment.

Keratinocyte: a type of epithelial cell that forms the epidermis; these cells proliferate in the stratum basale and differentiate to form the upper layers of the epidermis

3. EPIDERMAL TIGHT JUNCTIONS: KEY COMPONENT OF SKIN BARRIER INTEGRITY

Multicellular organisms use epithelial sheets to separate their bodies from the external environment and their organs from one another. Epithelia serve as selectively permeable barriers for fluids that differ in their chemical compositions. TJs are specialized intercellular adhesion complexes that are crucial for limiting molecular movement through the paracellular pathway because they seal the intercellular spaces of epithelial cellular sheets (16). TJs consist of TJ strands, which are composed mainly of four transmembrane claudin proteins on the plasma membrane that function in a structurally similar manner to zip locks to seal the adjacent plasma membrane (17, 18). Other major components of TJs include the transmembrane proteins occludin and tricellulin and the intracellular scaffold protein ZO-1 (19).

TJs form between keratinocytes in the second SG layer (SG2 cells) in the epidermis of mice and humans (**Figure 1b**) (20–22). In vivo studies demonstrated that intradermally injected tracer molecules of 557 Da, 1,500 Da, 5 kDa, and 32 kDa in size were halted at the TJ structures of healthy mouse skin. However, in mice with hapten-induced dermatitis, mRNA expression of TJ and SC components markedly decreased, and TJ barrier function was impaired in a size-dependent manner: The TJ leaked small tracers (<5 kDa) but not large tracers (>30 kDa). These results suggest that skin inflammation induces the development of leaky TJs (22). Skin inflammation also impairs TJ and SC barrier function, enhancing skin's permeability to external antigens and leading to a vicious cycle of skin barrier dysfunction and inflammation. Indeed, decreased claudin-1 expression induces skin inflammation and ichthyosis, a keratotic disorder characterized by a dry, thickened SC in mice (23). Furthermore, the absence of claudin-1, a major claudin in the epidermis, causes leaks in the epidermal TJ barrier and defective SC formation, resulting in early neonatal death in mice, likely due to water loss from the body surface (20). In humans, claudin-1 null mutations result in neonatal ichthyosis-sclerosing cholangitis syndrome (24) (**Table 1**). Collectively, these observations indicate that TJs play vital roles in both sealing the spaces between keratinocytes and maintaining the integrity of the skin barrier (11). These functions extend beyond the interfollicular epidermis to encompass more complex three-dimensional structures such as hair follicles and sebaceous glands (25).

3.1. Coordination of Tight Junctions and Epidermal Langerhans Cells

LCs are dendritic antigen-presenting cells found in the epidermis (26). These cells play a crucial role in the body's immune response. They are primarily involved in detecting, capturing, and presenting antigens. LCs remain below TJs and project their dendrites upward in their steady state. SC barrier perturbation, such as tape stripping or acetone treatment, stimulates the production of cytokines, including TNF- α and IL-1 β , in the epidermis and activates LCs (27, 28). Once activated, LCs express high levels of MHC-II, extend their dendrites beyond the TJ barrier, and capture external antigens that penetrate the SC in both human epidermis and mouse epidermis (21, 22, 29) (**Figure 1c**; **Supplemental Video 1**). During this process, new TJs are formed between the LC dendrites and the keratinocytes of the SG2 layer, enabling the uptake of external antigens without disturbing the barrier function of the TJs. The activated LCs then migrate out of the epidermis, present the captured antigens to T cells in the draining lymph nodes, and induce a T helper 2 (Th2) immune response with IgG1 antibodies against the captured antigens (30).

In response to skin barrier disruption and exposure to allergens and pathogens, keratinocytes release chemokines (e.g., CCL1, CCL2, CCL5, CCL11, CCL13, CCL17, CCL18, CCL20, CCL26, and CCL27) and alarmins [type 2 cytokines such as thymic stromal lymphopoietin

Occludin:

a four-transmembrane protein that anchors and regulates the permeability of tight junctions

Tricellulin:

a four-transmembrane protein that closes the gap of tricellular tight junctions, which seal the intercellular space at tricellular contacts

ZO-1: a cytosolic peripheral membrane protein that binds to proteins such as claudins and occludin, thereby acting as a scaffold for the tight junction

Supplemental Material >

(TSLP), IL-18, IL-25, and IL-33], promoting the infiltration of immune cells [i.e., T cells; innate lymphoid cells (ILCs), group 2 (ILC2s); eosinophils; mast cells; basophils; monocytes; LCs; and dendritic cells (DCs)] in the skin (31, 32). These alarmins subsequently induce ILC2s, Th2 cells, and basophils to release type 2 cytokines (32–36). In vitro studies using human epidermal equivalents have shown that IL-4 and IL-13 reduce filaggrin (a key SC component) and claudin-1 protein expression (37, 38), further promoting SC and TJ disruption, increased permeability, and allergen and pathogen penetration. Notably, whole-mount epidermal sheets from healthy individuals and from patients with AD demonstrated a fivefold increase in the number of LCs with TJ penetration in erythematous lesional skin of patients with AD compared with nonlesional skin of patients with AD or the skin of healthy individuals, suggesting that antigen capture ability also increases in AD (29). These findings underscore the SC's pivotal role as a primary barrier in the control of cutaneous sensitization. Serving as the initial line of defense, the SC regulates the penetration of external allergens and pathogens into the epidermis. Thus, when the integrity of the SC barrier is compromised—whether by genetic factors such as filaggrin deficiency or by external influences like low humidity or excessive washing with soap—external antigens can more readily penetrate the SC (29).

3.2. Coordination of Tight Junctions and Epidermal Nerves

Epidermal sensory nerves play crucial roles in sensory perception and transmit signals to the brain in response to external stimuli. Epidermal nerve endings frequently extend and retract while remaining under TJs in a steady state (39). In normal skin, epidermal nerve endings undergo local pruning at intersections with newly formed TJs, which causes epidermal nerve endings to remain below the TJs. In contrast, this pruning process is impaired in proinflammatory conditions, and nerve endings remain extended outside the TJ, inducing Ca^{2+} signaling at nerve endings (39) (**Figure 1c**). This circumstance suggests that disruption of homeostatic pruning of epidermal nerve endings may lead to pathological itching. These findings, along with those related to LCs, indicate that TJs are pivotal control centers that maintain the integrity and homeostasis of the skin barrier.

3.3. Shape Matters in Structural Dynamics and Barrier Function of the Epidermis

The shape of keratinocytes has long been perceived as hexagonal prisms (40). However, en face views of epidermal TJs have demonstrated that double-edged polygons are found in approximately 1 of 10 polygons (21). In addition, intravital imaging of epidermal TJs has demonstrated that smaller TJ edges are always new and located below the old edges (13). However, these observations were not well explained by the conventional hexagonal prism model. These conflicting findings led us to consider an f-TKD (13).

Kelvin's tetrakaidecahedron is a geometric solid originally proposed by the Scottish mathematician and physicist Sir William Thomson (later Lord Kelvin) in 1887 (41). It is a polyhedron composed of 14 faces: 6 square faces and 8 hexagonal faces. Kelvin proposed this shape in an attempt to solve the problem of dividing space into equal volumes with the smallest surface areas, which is important for understanding the efficiency of natural and manufactured structures.

TJs form at the edges of f-TKD cells, and only a single layer of cells (SG2 layer cells) form TJs at the apical edges of lateral cell-cell contacts (**Figure 1b**; **Supplemental Video 2**). When a TJ-forming cell (old cell) is replaced by a new cell located immediately beneath it, the new cell first forms a new TJ polygon beneath the old cell. Therefore, each TJ polygon becomes temporarily double-edged during cell replacement. Subsequently, the TJ polygon of the old cell disappears, and the old cell is eventually extruded outside the TJ (**Figure 1b**; **Supplemental Video 3**). The

temporary formation of the double-edged TJ polygon at each cell turnover site maintains the homeostasis of the TJ. The f-TKD shape helps explain how cells not only can pack efficiently, minimizing the gaps between them, but also restore TJ function, constantly differentiating to maintain the skin's barrier integrity.

4. STRATUM CORNEUM: THE FRONT LINE OF SKIN BARRIER DEFENSE

To adapt to terrestrial life, vertebrates have developed an air-liquid interface barrier, the SC, as the outermost layer of the epidermis. The SC protects against pathogens, invasion by allergens, and dehydration (42, 43). Formation of the SC involves a complex, time-dependent process known as cornification, which begins in the SG. Cornification consists of (a) programmed cell death of SG1 cells, termed corneoptosis (12), which degrades the cells' organelles and results in the formation of corneocytes; (b) secretion of the contents of the lamellar body to form intercellular lipid lamellae (10, 44); (c) covalent binding of acylceramides to proteins of the cornified cell envelope (CCE) to form the corneocyte lipid envelope (CLE), which serves as the scaffold for intercellular lipid lamellae (44–46); (d) cross-linking of proteins by transglutaminases at the cell periphery to wrap the corneocytes with the CLE and CCE (10, 47); and (e) alteration of the cytoskeleton to form a dense intracellular keratin network (44, 46).

The SC is unique in its function as a skin barrier, despite being composed of dead keratinocytes (corneocytes) that do not exhibit energy-dependent activities, such as ATP-driven active transport (48, 49). The SC maintains site-specific thickness by continuously shedding corneocytes through a process termed desquamation. Furthermore, the SC hosts the skin microbiota, which plays an essential role in priming both the innate and adaptive skin immune systems. These characteristics contribute to overall SC homeostasis.

In inflammatory skin diseases, such as AD, affected keratinocytes may retain nuclei within the SC, a condition known as parakeratosis. Such diseases often result in a thicker SC (hyperkeratosis) compared with that of healthy skin and in dysbiosis of the skin microbiota. Therefore, a deeper understanding of the processes involved in cornification and SC homeostasis is essential for developing new treatment strategies for inflammatory skin diseases.

4.1. Corneoptosis: The Distinctive Cell Death Process in SG1 Cells

Corneoptosis is distinguished by the complete degradation of organelles during cell death, and it occurs without inducing inflammation. Unlike the cellular remnants of apoptosis or necrosis, which are typically cleared by macrophages or adjacent cells through efferocytosis, corneocytes—the remnants of SG1 cells—are integral in forming the SC (12).

Intravital imaging studies utilizing mice that express two intracellular Ca^{2+} probes within the epidermis—one sensitive to pH and the other less sensitive to pH—have revealed key insights into the dynamics of corneoptosis (12). This cell death process begins with a prolonged elevation of intracellular Ca^{2+} concentration lasting approximately 60 min (corneoptosis phase I), which is then followed by rapid intracellular acidification (corneoptosis phase II) (Figure 2a; Supplemental Video 4). During corneoptosis phase II, organelles such as keratohyalin granules (KHGs), the nucleus, and mitochondria degrade without triggering inflammation. KHGs are liquid-like, membraneless organelles that contain the histidine-rich protein profilaggrin, a precursor of filaggrin in which 10–12 filaggrin monomers are linked. Through liquid-liquid phase separation, profilaggrin drives the formation of KHGs and increases their viscosity. A network of keratin 1 and keratin 10 (K1/K10) filaments binds to profilaggrin, resulting in the formation of a stiff cytoplasmic web that restricts the coalescence of KHGs, thereby stabilizing the structure of SG1 cells (50). The

Acylceramide:

differs from normal ceramides because it has an additional hydrophobic chain (linoleic acid) alongside the two hydrophobic chains (sphingoid base and fatty acid) common to ceramides

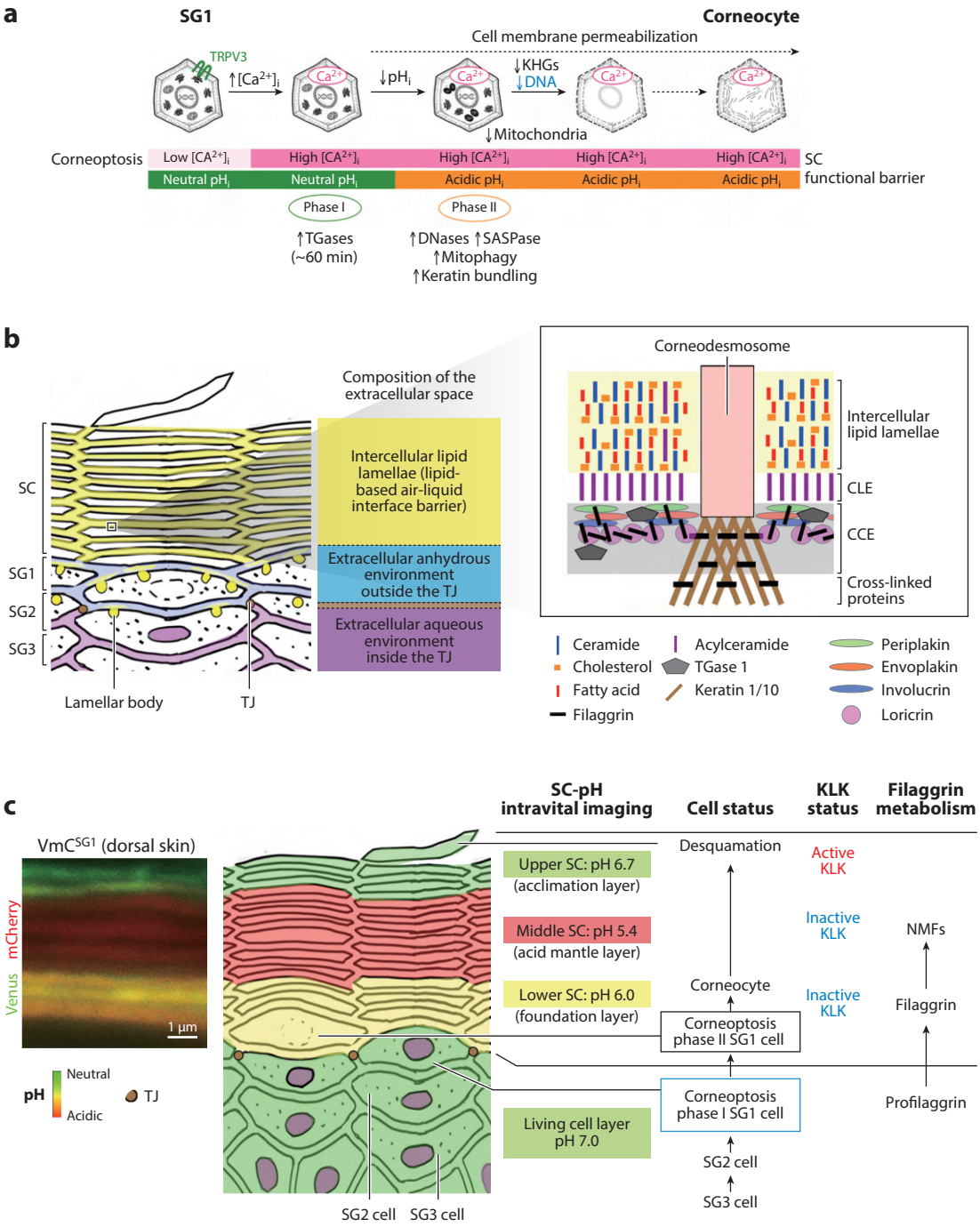
Transglutaminase: an enzyme that catalyzes the formation of an ϵ (γ -glutamyl)-lysine cross-link

Keratin: a type of intermediate filament that forms the cytoskeleton; during cornification, keratin 1/10 bundles are loosened and reassembled by filaggrin

Profilaggrin:

precursor of the filaggrin monomer, constituting 10–12 repeats of filaggrin monomers linked and stored in keratohyalin granules

Supplemental Material >



(Caption appears on following page)

Figure 2 (Figure appears on preceding page)

Corneoptosis and structure of the SC. (a) Schematic illustration of corneoptosis. During corneoptosis, corneocytes undergo two sequential states: In phase I, which lasts for approximately 60 min, the combination of high $[Ca^{2+}]_i$ and neutral pH_i activates TGases. In phase II, the combination of high $[Ca^{2+}]_i$ and acidic pH_i induces DNase (i.e., DNase1L2 and DNase2) activation, SASPase activation, mitophagy, and keratin network alterations. As a result, KHGs and mitochondria are eliminated, and DNA is completely degraded. The SG1 cell death process is then complete, and corneocytes are formed. Panel adapted from Reference 12 (CC BY 4.0). (b) Basic structure of the upper epidermis (*left*). Intercellular spaces are filled with lipids in the SC and with water under the SC. The TJ barrier is considered to divide the extracellular environment into two parts: the extracellular space outside the TJ barrier (*light blue*) and inside the TJ barrier (*purple*). (*Right*) Basic structure of the SC (region zoomed in from *box* on the *left*). The SC is composed of corneocytes and intercellular lipid lamellae. The surface of the corneocytes is coated with a CLE, a thin single layer of protein-bound ceramides mainly derived from acylceramide. A CCE is formed beneath the CLE via CCE protein (e.g., envoplakin, periplakin, and involucrin) cross-linking. These proteins are cross-linked beneath the corneocyte plasma membrane by TGase 1. Filaggrin functions as a matrix protein that packs keratin filaments. The keratin-filaggrin bundle is a fundamental structure in corneocytes. In addition, filaggrin is cross-linked with CCE proteins. The corneodesmosome, which is distributed on the corneocyte surface and interacts with cadherin proteins, exists in the extracellular space of adjacent corneocytes. Furthermore, the corneodesmosome interacts with keratin-filaggrin bundles in the intracellular spaces of the corneocytes. (c) Representative confocal microscopic images of the SC-pH zones (*left*) of the dorsal skin of VmC^{SG1} mice (hairless animals expressing pH probe Venus-mCherry in the SG1 layer). Image depicting the Venus and mCherry expression in mice adapted from Reference 49 (CC BY 4.0). (*Right*) The function of the three stepwise SC-pH zones. Corneocytes undergo acidification, thereby inducing three-tiered pH zonation and generating the moderately acidic lower zone (pH 6.0), the acidic middle zone (pH 5.4), and the nearly neutral upper zone (pH 6.7). The lower SC-pH zone results from acidification during corneoptosis phase II. Moreover, this zone corresponds to the region where filaggrin localizes and degrades to become NMFs in the middle SC-pH zone. Abbreviations: $[Ca^{2+}]_i$, intracellular Ca^{2+} concentration; CCE, cornified cell envelope; CLE, cornified lipid envelope; KHGs, keratohyalin granules; KLK, kallikrein-related peptidase; NMFs, natural moisturizing factors; pH_i, intracellular pH; SASPase, skin aspartic protease; SC stratum corneum; SG, stratum granulosum; TGase, transglutaminase; TJ, tight junction; TRPV3, transient receptor potential vanilloid 3.

physiological acid dissociation constant of histidine is approximately 6.1 (51), and studies using a human keratinocyte cell line with a fluorescent phase separation probe have shown that when pH shifts from 7.4 to 6.2, the dissolution of KHGs commences, causing profilaggrin to transition from a granular-like state to a more diffuse cytoplasmic distribution (50). These results imply that the transition of SG1 cells to corneocytes critically depends on this acidification process.

Indeed, *in vitro* experiments using SG1 cells have demonstrated that both high $[Ca^{2+}]$ and acidic pH (conditions similar to those seen in corneoptosis phase II) are necessary for the degradation of KHGs and of nuclear DNA (12) (**Supplemental Video 5**). Critical to this degradation process is SASPase (skin aspartic protease), which cleaves profilaggrin to filaggrin, and nucleases DNase1L2 and DNase2, which show optimal activity at acidic pH levels and are active in SG1 cells (52–55). The rapid drop in pH during phase II facilitates the action of these enzymes, leading to efficient organelle breakdown. This finding is supported by evidence showing defective degradation of nuclear DNA in mouse skin that lacks *Dnase1l2* and *Dnase2* genes (55). Additionally, live imaging of organotypic cultures of human epidermis demonstrated that keratinocytes in the upper epidermis upregulate mitochondrion-localized autophagy receptor NIX and induce mitophagy via the GTPase DRP1 (dynamin-related protein 1), thereby ensuring mitochondrial degradation and conversion of SG1 cells to corneocytes (56) (**Figure 2a**).

4.2. Structure of the Stratum Corneum

The SC has four major components, arranged from the inside of corneocytes to the outside: keratin/filaggrin and their degradation products, which fill the cytoplasm of cornified cells; the CCE; the CLE; and the lipid lamellae (57, 58) (**Figure 2b**).

Lipid lamellae are multilayered structures that occupy the intercellular spaces of corneocytes and are composed of ceramides, cholesterol, and free fatty acids. In addition, they incorporate proteins such as hydrolytic enzymes, which modify lipids; corneodesmosin, which modifies

Supplemental Material >

Involucrin: a protein that acts as a scaffold for the cornified cell envelope and cross-links with envoplakin, periplakin, and loricrin via transglutaminase

Loricrin: a protein that constitutes up to 80% of the cornified cell envelope protein

desmosomes into corneodesmosomes; proteases and protease inhibitors, which control desquamation; and antimicrobial peptides, which eliminate pathogenic microorganisms (59). These lipids and proteins are synthesized in the SG, stored in lamellar bodies, and secreted into the extracellular space during the transition to the SC. This secretion process thereby develops the lipid lamellae and CLE that ensure barrier function and SC homeostasis (60).

The corneodesmosome, a specialized desmosome apparatus distributed on the surface of corneocytes, consists of the cadherin proteins desmoglein 1 and desmocollin 1—along with plakoglobin, plakophilin, envoplakin, periplakin, and corneodesmosin—and is essential for adhesion between corneocytes and for SC barrier function (44). Corneodesmosin interacts with cadherin proteins, exists in the extracellular space of adjacent corneocytes, and supports corneocyte-corneocyte adhesion (47) (**Figure 2b**). Patients who are deficient in corneodesmosin and desmoglein 1 have peeling skin syndrome 1 (61, 62) and SAM (severe dermatitis, multiple allergies, and metabolic wasting) syndrome (62, 63), respectively (**Table 1**). Because both conditions manifest detachment of the SC layer from the epidermis, the epidermis of these patients shows increased permeability to allergens and enhanced percutaneous immunization, which lead to the development of severe dermatitis, food allergies, and increased serum IgE levels (61–63).

Ceramides are the major lipid species in the SC and account for approximately 50% of SC lipids. Epidermis-specific ceramides, such as acylceramides, account for 18% of SC ceramides (64–66). The CLE consists of a single, thin layer that covers corneocytes and is mainly composed of CerEOS (acylceramide composed of esterified ω -hydroxy fatty acids and sphingosines). The lipoxygenases ALOX12B and ALOXE3 oxidize the linoleic acid moiety of CerEOS, and SDR9C7 dehydrogenates the oxidized linoleic acid in acylceramide to promote the binding of CerEOS to the CCE proteins (44, 46). Thus, loss-of-function mutations in genes involved in acylceramide synthesis (e.g., *ALOX12B*, *ALOXE3*, and *SDR9C7*) cause reduced CLE formation, increased SC permeability, impaired desquamation, and the development of autosomal recessive congenital ichthyosis, an inherited keratotic disorder characterized by a dry, thickened SC (hyperkeratosis) (44, 45, 67–73) (**Table 1**). Notably, alteration in ceramide composition and decreased acylceramide levels in the SC have been implicated in inflammatory and allergic properties, as well as in impaired barrier function in patients with AD (74, 75). In addition to the alteration of ceramide production in the keratinocytes by IFN- γ , TNF- α , and Th2 cytokines, disruption of the ceramide profile and of barrier function itself leads to the continuous production of cytokines and chemokines such as IL-1 α , TNF- α , and CCL17, which exacerbate inflammation in AD (76, 77). Transmembrane protein 79 (Tmem79), also known as matrin, is involved in the secretion of lamellar body contents (78, 79). Tmem79-deficient mice exhibit spontaneous dermatitis with impaired SC formation, increased production of abnormal sebaceous lipids, and dysbiosis of skin microbiota (80, 81). A meta-analysis of patients with AD indicated that a missense mutation in the *TMEM79* gene had a small but significant association with AD (60, 79) (**Table 1**). These results suggest that the contents of the lamellar body are essential for normal cornification and SC barrier function.

The CCE, composed of proteins such as involucrin, loricrin, and profilaggrin, is the most insoluble component of the SC. The proteins are cross-linked beneath the plasma membrane of corneocytes by transglutaminase 1, and defects in transglutaminase 1 can cause autosomal recessive congenital ichthyosis (82) (**Table 1**). Filaggrin functions primarily as a matrix protein that packs keratin filaments, and this keratin-filaggrin bundle is a fundamental structure in corneocytes; however, small amounts of filaggrin also cross-link with CCE proteins. Notably, type 2 cytokines IL-4 and IL-13 can reduce the protein expression of involucrin, loricrin, and FLG (83, 84). In addition, skin inflammation mediated by IL-31, TSLP, IL-17A-producing CD4⁺ T (Th17) cells (via IL-17), Th22 cells (via IL-22), and Th1 (IFN- γ -producing CD4⁺ T) cells (via IL-1 α , IL-1 β , and

TNF- α)—as well as by repetitive scratching, detergent use, and low humidity—can also reduce FLG expression (77, 85–92). Thus, as the initial line of defense, the SC regulates the penetration of external allergens and pathogens into the epidermis, playing a crucial role in determining the degree of cutaneous sensitization.

The SC is formed by the development of the abovementioned structures through cornification. Various types of genetic ichthyosis, long regarded solely as a structural disease of impaired cornification, have also been shown to involve the activation of the Th17 pathway and innate immunity (93–96). These findings suggest that biologics for targeting the Th17 pathway might offer new therapeutic approaches for treating certain types of congenital ichthyosis, as evidenced by the reduction of erythema and scaling in patients treated with the anti-IL-17A antibody secukinumab (97, 98).

4.3. Three Stepwise pH Progressions in the Stratum Corneum

Traditionally, the SC, with its covalently cross-linked and insoluble proteins, has been difficult to analyze using conventional methods in cell biology, leading to perceptions of the SC as a homogeneous barrier (58). However, studies using advanced imaging techniques such as scanning electron microscopy (SEM) (99) and time-of-flight secondary ion mass spectrometry (TOF-SIMS) (100) have unveiled the SC's complex stratification. These studies identified three distinct layers within the SC—upper, middle, and lower—each characterized by unique water penetration properties and binding properties. SEM revealed differential swelling behaviors when the SC layers are soaked in water: The lower SC swells; the middle SC remains largely unaffected, maintaining its thickness; and the upper SC experiences significant swelling, loosening its intracellular filament packing (99). TOF-SIMS, which can visualize the distribution of natural substances without staining, highlighted a concentration of arginine, a major component of filaggrin-derived natural moisturizing factors (NMFs), in the middle SC, underscoring arginine's crucial role in skin hydration. A soaking assay using TOF-SIMS revealed that the upper SC allowed the influx and efflux of various metal ions, whereas the middle and lower layers exhibited distinct barrier properties (100). Furthermore, analysis by cryo-electron microscopy in combination with molecular dynamics simulation revealed that the lipid lamellae show different lamellar patterns between the first and second corneocytes, between the second and third corneocytes, and above the third corneocytes (101, 102). Collectively, these results imply that the SC is not a homogeneous layer but rather exhibits differentiation.

Recently, intravital imaging studies in mice expressing pH probes in corneocytes demonstrated that corneocytes undergo differentiation, creating a three-tiered pH zonation (49). This zonation consists of a moderately acidic lower zone (pH 6.0), an acidic middle zone (pH 5.4), and a nearly neutral upper zone (pH 6.7) (**Figure 2c**; **Supplemental Video 6**). The lower SC-pH zone, originating from acidification during corneoptosis phase II, corresponds to the zone wherein filaggrin exists as monomers (103). The middle SC-pH zone functions as the acid mantle, providing a protective barrier against external bacteria, and corresponds to the area wherein filaggrin breaks down into NMFs (60). The upper SC-pH zone is adaptable, adjusting its pH in response to external conditions, and is influenced by the skin microbiota (49).

Desquamation, the shedding process at the outermost corneocyte layer that maintains specific SC thickness, involves the degradation of corneodesmosomes by kallikrein-related peptidases (KLKs): KLK5, KLK7, and KLK14 (47, 104). The catalytic activity of these enzymes is pH dependent, with optimal activity occurring at a neutral pH, and their catalytic activity is also regulated by coexpressed proteinase inhibitors such as lymphoepithelial Kazal-type-related inhibitor (LEKTI) (105–107). A mathematical model that incorporated the three-tiered zonation simulated the high catalytic activity of KLKs in the SC upper layer, promoting effective corneocyte

Natural moisturizing factor (NMF):

a component of corneocytes for moisturization; it is mainly produced by filaggrin degradation and is composed of amino acids and their derivatives

Supplemental Material >

shedding in desquamation (49). In contrast, mice lacking claudin-1, an essential transmembrane component of TJs, in the epidermis develop spontaneous dermatitis with a dry, thickened SC (23) and do not exhibit this stepwise SC-pH zonation, displaying a uniformly neutral pH across the SC. This claudin-1 deficiency affects intercellular lipid composition (108) and leads to the accumulation of prematurely processed filaggrin, suppression of subsequent production of NMFs, and impaired desquamation (49). These findings suggest that the three-tiered pH zonation is crucial for maintaining SC thickness and skin barrier function, thereby contributing to overall SC homeostasis.

Patients with Netherton syndrome, a monogenic skin disease resulting from the loss-of-function mutation of *LEKTI*, display high catalytic activity of KLKs; extensive desquamation; activation of the Th17 and Th2 pathways; downregulation of claudin-1 expression; reduction of sphingolipid and ceramide levels, thereby causing hair shaft abnormalities, AD-like dermatitis, food allergies, and asthma; and markedly elevated serum IgE levels (62, 96, 109, 110) (**Table 1**). Similarly, AD downregulates NMF, filaggrin, ceramides, and claudin-1 expression (23, 91). Furthermore, the intravital SC-pH imaging of mice that received MC903, the reagent that stimulates TSLP release from keratinocytes and thereby induces AD-like dermatitis, showed neutralization of the middle SC-pH zone (49). Thus, targeting the impaired stepwise SC-pH zonation may offer a new therapeutic approach for inflammatory skin disorders.

5. HAIR FOLLICLE: COMPLEX STRUCTURE ORCHESTRATING IMMUNE RESPONSES

5.1. Tight Junctions in the Hair Follicles, Sebaceous Glands, and Sweat Glands

The interfollicular epidermis of the mammalian epidermis consists of sweat glands, sebaceous glands, and hair follicles (HFs). Notably, sweat glands and sebaceous glands are delineated by epithelial cellular sheets lacking the SC. To maintain skin barrier homeostasis, the paracellular spaces of luminal cells of the eccrine sweat glands are sealed by TJs (111). These TJs, which predominantly contain claudin-3, play a crucial role in controlling the internal environment. In mice deficient in claudin-3, sweat leaks into the paracellular spaces of sweat gland cells, potentially leading to skin irritation (112). Similarly, a decrease in claudin-3 expression in sweat glands has been observed in AD lesions (112). Sweat leakage into the dermis may partly contribute to the tingling pain or itch frequently experienced by patients with AD (113).

HFs are complex structures consisting of three distinct segments: the lower segment (bulb and suprabulb), the middle segment (isthmus), and the upper segment (infundibulum) (114). The lower segment stretches from the follicle base to the arrector pili muscle insertion. The middle segment extends from the arrector pili muscle insertion to the opening of the sebaceous gland duct, and the upper segment reaches from the opening of the sebaceous duct to the follicular orifice. Sebaceous glands, composed of sebocytes, drain into the infundibulum of the hair follicle (**Figure 1a**). The TJs on the luminal side of the sebocyte surface are crucial for the differentiation of holocrine secretions (25). Knockout of claudin-1 in mice results in the accumulation of partially degraded sebocytes within the sebaceous ducts, indicating that a functional TJ barrier is essential for proper sebocyte differentiation and secretion (25).

The epithelium of the HF includes several morphologically distinct compartments such as the hair shaft, cuticle, Huxley's and Henle's layers (forming the inner root sheath), the companion cell layer, and the outer root sheath (ORS); the ORS is a stratified epithelium that wraps around the follicle and is contiguous with the epidermis (115, 116) (**Figure 1a**). In HFs in the anagen phase, a continuous TJ barrier is observed from the infundibulum down to the lower-central/upper-suprabulbar region. Specifically, in the infundibulum, the barrier is formed between SG2 cells. In

the isthmus, central, and upper-suprabulbar regions, the TJ barrier is localized in the companion cell layer of the ORS. An additional TJ barrier was observed between Henle's and Huxley's layers in the lower-central/upper-suprabulbar region. No barrier was observed in the bulbar region (117). These findings highlight the crucial role of TJs as the primary line of defense against external pathogens in skin appendages and suggest that compromised TJ function may contribute to the pathogenesis of AD through disrupted skin barrier integrity.

5.2. Hair Follicle as an Epicenter of Immune Activities

Increasing evidence underscores the critical role of HFs in the recruitment and positioning of skin-resident immune cells. Keratinocytes within the infundibulum produce the chemokine CCL20, which guides the entry of CCR6⁺ LCs into the epidermis following mechanical stress (118). Additionally, the CCL20-CCR6 axis facilitates the migration of T regulatory cells (Tregs) to the skin during the neonatal period, establishing immune tolerance to commensal microbes. This process is enhanced by skin commensal bacteria, which increase the production of CCL20 from the upper hair follicle and, consequently, increase the frequency of Tregs (119). Moreover, skin-resident Tregs express the Notch ligand Jagged-1 or glucocorticoid receptor to induce TGF- β 3, which activates HF stem cell proliferation and supports hair regeneration (120, 121).

The keratinocytes in the infundibulum and isthmus also produce survival cytokines IL-7 for skin-resident CD4⁺ memory T (skin CD4⁺ Trm) cells and IL-15 for skin-resident CD8⁺ memory T (skin CD8⁺ Trm) cells (122, 123). Blocking HF-derived IL-7 has been shown to decrease the population of cancerous skin CD4⁺ Trm cells in the epidermis of a mouse model of cutaneous T cell lymphoma (122). When IL-15 signaling was blocked by targeting CD122, a subunit of the IL-15 receptor on skin CD8⁺ Trm cells, melanocyte antigen-specific skin CD8⁺ Trm cells were reduced and durable repigmentation was facilitated in a mouse model of vitiligo, an autoimmune skin disease characterized by the destruction of melanocytes by autoreactive skin CD8⁺ T cells (124). These findings highlight the potential of targeting HF-derived cytokine production as a therapeutic strategy for various skin diseases, including cutaneous T cell lymphoma and inflammatory skin disease.

Skin-resident ILC3s, located within hair follicles in proximity to sebaceous glands, survive via keratinocyte-derived IL-7 and thymic stromal lymphopoietin, leveraging the CCL20-CCR6 axis. These ILC3s express TNF receptor ligands that limit the growth of sebocytes by repressing the Notch signaling pathway. A loss of skin-resident ILC3s results in sebaceous hyperplasia, increased production of antimicrobial lipids, and restricted growth of gram-positive bacterial communities (125, 126). Furthermore, the expression of the transmembrane endopeptidase ADAM10 in the upper HFs is crucial for regulating the skin microbiota. Disruption of the ADAM10-Notch signaling axis leads to dysbiosis with prominent *Corynebacterium* species and triggers ILC2-mediated inflammation, which depends on IL-7R, S1P1R, and CCR6 and results in pyroptotic cell death of HFs, causing irreversible alopecia (127). Collectively, these results underscore the multifaceted role of HFs in the regulation of immune cells associated with both innate and acquired immunity and in the facilitation of host-microbe symbiosis.

6. SKIN-RESIDENT MEMORY T CELLS: SKIN IMMUNE SENTINELS

Memory T cells are classified into two categories based on their migratory behavior: tissue-resident memory T (Trm) cells, which dwell in the periphery and were introduced in Section 5.2, and circulating memory T cells, which are mainly found in the blood and in lymphoid organs (128–130). Circulating memory T cells can be divided into two main types: central memory T cells, which preferentially home to secondary lymphoid organs, and effector memory T cells,

Vitiligo:

an autoimmune skin disease characterized by patchy depigmentation resulting from the destruction of melanocytes by autoreactive CD8⁺ T cells

which are capable of traversing nonlymphoid tissues (131). Trm cells establish long-term residency within peripheral tissues to mediate local immunity against invading pathogens, allergens, and burgeoning cancers (131, 132). Trm cells develop in response to infection, toxin exposure, allergies, and autoimmune diseases (133), and they enhance immune responses upon re-recognition of antigens (131, 134). Importantly, Trm cells not only accumulate at the site of primary inflammation but also distribute widely to distant, previously uninvolved skin, enabling broader host protection against secondary challenges (135). Surprisingly, healthy human skin contains approximately 20 billion T cells, twice the number found in the blood (136). Most T cells in healthy human skin are CD69⁺CD103⁺ or CD69⁺CD103⁻ skin Trm cells, accounting for 62% and 27% of T cells in the epidermis and for 30% and 49% of T cells in the dermis, respectively (137).

The key feature of Trm cells is the shutdown of sphingosine-1-phosphate (S1P) receptor 1 (*S1PR1*) and receptor 5 (*S1PR5*) genes, which facilitate tissue egress, coupled with the upregulation of tissue-specific molecules that mediate T cell retention (131, 138–140). In the epidermis, effector memory T cells recruited by viral infection depend on TGF- β to differentiate into skin Trm cells (141, 142). TGF- β activation is mediated exclusively via the integrins α v β 6 and α v β 8, which are expressed by keratinocytes (143–145). The exposure of T cells to activated TGF- β efficiently induces the expression of α E integrin, which complexes with the integrin β 7 to form α E β 7 (also called CD103) and binds to E-cadherin expressed by keratinocytes (141, 146, 147). Although the expression of CD103 in Trm cells is not universal across all tissues, Trm precursors in the skin induce CD103 expression upon the entry of CD103 to the epidermis, along with CD69 expression, which antagonizes S1PR1 activity and thereby enforces tissue retention (141, 146). Deletion of CD103 or CD69 on virus-specific T cells leads to a significant reduction in the number of skin Trm cells following virus infection, suggesting that this upregulation of CD103 and CD69 on skin Trm cells is important for their development and survival (141). In addition, hair follicle-derived IL-15 and IL-7 are important for the survival of skin CD8⁺ Trm cells and CD4⁺ Trm cells, respectively (122). Furthermore, skin CD8⁺ Trm cells upregulate fatty acid-binding protein 4 (FABP4) and FABP5, thereby increasing mitochondrial oxidative metabolism in the presence of exogenous free fatty acids, which supports long-term survival of skin CD8⁺ Trm cells (148, 149). Collectively, these results suggest that skin Trm cell survival is regulated by intraepidermal- and interepidermal-derived (*a*) cytokines, (*b*) chemokines, and (*c*) exogenous free fatty acids.

Growing evidence supports the finding that skin CD8⁺ Trm cells are heterogeneous. In addition to Trm type 1 (Trm1) cells—skin CD8⁺ Trm cells that produce IFN- γ —there are Trm type 17 (Trm17) cells—Trm cell subsets that produce IL-17 and differentially contribute to immune responses. Functional assays have revealed that Trm1 cells are associated with viral and tumor control (150), whereas skin Trm17 cells promote bacterial defense and wound healing (151–153). Moreover, Trm1 cells and Trm17 cells differentially contribute to skin pathologies, including vitiligo and psoriasis, respectively, wherein they persist long after disease resolution (124, 150, 154–156). In human skin, CD8⁺CD49a⁺ Trm cells produce IFN- γ (Trm1 cell phenotype), whereas CD8⁺CD49a⁻ Trm cells produce IL-17 (Trm17 cell phenotype). Additionally, CD8⁺CD49a⁺ Trm cells from healthy skin rapidly induce the expression of the effector molecules perforin and granzyme B when stimulated with IL-15, thereby promoting a strong cytotoxic response. Thus, CD49a expression delineates CD8⁺ Trm cell specialization in the human skin barrier and correlates with the effector cell balance found in distinct inflammatory skin diseases (150). Furthermore, a recent study demonstrated that Trm1 cells depend on a T-bet–Hobit–IL-15 axis, whereas Trm17 cells develop independently of these factors. Instead, c-Maf commands a tissue-resident program in Trm17 cells parallel to that induced by Hobit in Trm1 cells, with the ICOS–c-Maf–IL-17 axis (157). Accordingly, the targeting of this pathway can ablate skin Trm17 cells without compromising their Trm1 cell counterparts. Collectively, these results indicate that skin Trm

cells rely on distinct molecular circuitries, which can be exploited to modulate local immunity strategically.

7. SKIN MICROBIOME: BALANCING PROTECTION AND INFLAMMATION IN THE STRATUM CORNEUM MICROENVIRONMENT

7.1. The Skin Microbiome as a Component of Barrier Function

The skin hosts a unique consortium of microorganisms, highlighting its critical role in barrier function. This microbiota, which consists of bacteria, fungi, viruses, and parasites, resides not only on the skin but also on other mucosal surfaces connected to the external environment, including the gastrointestinal tract, oral and nasal cavities, the respiratory tract, and the urinary tract. Beyond its role in maintaining human health, the microbiome has been implicated in a variety of systemic and local diseases, such as inflammatory bowel disease, diabetes, autism, and cancer (158–162).

The skin microbiota is essential for maintaining skin homeostasis and regulates immune responses and metabolic processes across various skin structures. It also guards against the invasion of pathogens from the environment. The composition of the skin microbiota varies significantly among individuals, and even within individuals, influenced by factors such as anatomical location, the density of skin appendages, age, and environmental conditions such as lipid composition, salinity, pH, peptide levels, and moisture (163), resulting in a rich diversity of microbiome profiles across different skin sites (164, 165).

Typically, the skin microbiota is categorized based on the characteristics of the skin at various body sites: seborrheic (oily) areas rich in sebaceous glands, such as the forehead and chest; moist areas abundant in sweat glands, such as the elbow fossae and knee fossae; dry areas, such as the forearms and buttocks; and moist and highly diverse areas, such as the feet, particularly the toe web spaces (166–168). These microorganisms are grouped into two main categories: a core microbiota that stably colonizes the skin and a transient microbiota, which is typically acquired from the external environment but dissipates within a period of hours to days (169). The core microbiota remains stable over long periods despite continuous exposure to external elements (163) and is considered to coexist in a symbiotic relationship that generally is harmless to the host. In contrast, transiently detected skin microorganisms under normal conditions are typically nonpathogenic (170).

7.2. Host-Microbe Interactions for Homeostasis in the Skin

While many interactions between the host and individual skin microorganisms are yet to be fully understood, the collective function of the skin microbiota is crucial for maintaining skin homeostasis. The skin microbiota plays diverse roles, both as a consortium and as individual species (**Figure 3a**). Understanding these interactions, both host-microbe and microbe-microbe, is vital for comprehending the broader function of the microbiota in maintaining skin equilibrium.

For instance, proteases produced by commensal skin bacteria are involved in the desquamation process of the SC, and the breakdown of sebum and free fatty acids by the microbiota contributes to the regulation of pH on the skin surface (171, 172). Additionally, the skin microbiota acts as a protective shield, preventing the invasion and establishment of potential pathogens by competing with the pathogens for habitat and nutrients, by producing antimicrobial peptides, and by activating immune responses. Host-microbe cross talk enhances skin barrier homeostasis through various pathways, including the maturation of barrier function via activation of aryl hydrocarbon receptors in keratinocytes (173), education of the immune system (174, 175), support of regeneration through IL-1R/MyD88 signaling in keratinocytes (176), facilitation of wound repair (177), and inhibition of the development of skin cancer (178). Furthermore, the skin microbiota promotes

Microbiome:
the entirety of
microorganisms and
their collective genetic
material present in a
particular environment

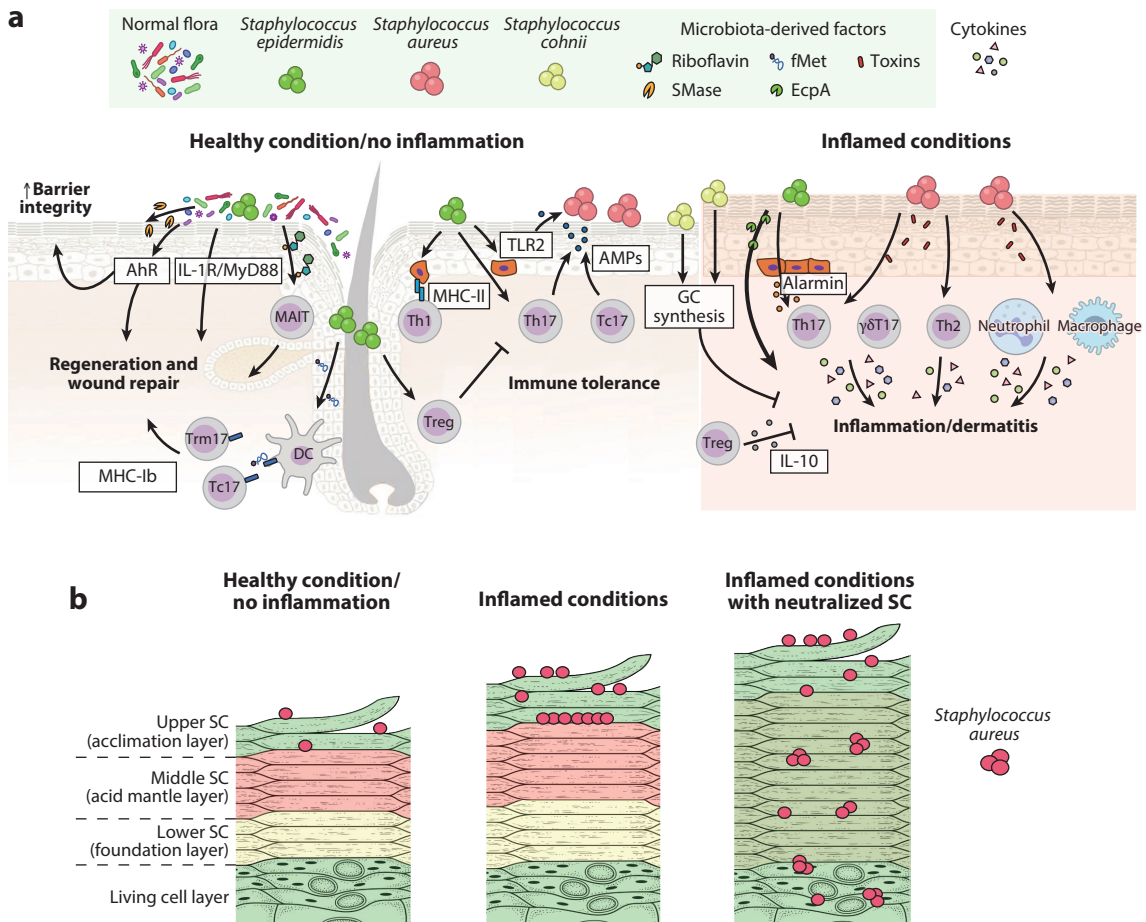


Figure 3

Immunological cross talk between host and skin microbiota. (a) In homeostatic conditions, normal flora, including *Staphylococcus epidermidis*, improves skin health by stimulating aryl hydrocarbon receptor (AhR) and secreting sphingomyelinase (SMase). Normal flora also produces riboflavin, which induces mucosal-associated invariant T (MAIT) cells, resulting in wound repair. Nonclassical MHC-Ib-restricted IL-17A-producing CD8⁺ T (Tc17) cells and tissue-resident memory type 17 (Trm17) cells are induced by *S. epidermidis*-derived *N*-formylmethionine (fMet) peptide ligands via dermal dendritic cells (DCs), promoting regeneration and tissue repair. *S. epidermidis* accumulates IFN- γ -producing CD4⁺ T (Th1) cells through keratinocyte-intrinsic MHC-II expression. Epidermal keratinocytes are stimulated to produce antimicrobial peptides (AMPs) against pathogens such as *Staphylococcus aureus* via Toll-like receptor 2 (TLR2) signaling, Tc17, and IL-17A-producing CD4⁺ T (Th17) cells, which are all induced by *S. epidermidis*. Commensal-specific T regulatory cells (Tregs) are induced in neonates, resulting in immune tolerance against skin microbiota. Colonization of *Staphylococcus cohnii* promotes glucocorticoid (GC) synthesis, suppressing inflammation in the skin. In inflamed conditions, the *S. epidermidis*-derived protease EcpA increases inflammatory responses. Keratinocytes stimulated by *S. epidermidis* produce alarmins such as IL-18, leading to Th17 cell accumulation. *S. aureus*-derived toxins such as δ -toxin, α -toxin, and PSM α (phenol-soluble modulin α) induce Th2, Th17, and IL-17A-producing $\gamma\delta$ T ($\gamma\delta$ T17) cells, producing inflammatory cytokines that amplify inflammation and dermatitis. These toxins and *S. aureus* invasion induce the accumulation of neutrophils and macrophages in the skin. Tregs suppress inflammatory responses by producing anti-inflammatory cytokines such as IL-10. (b) *S. aureus* is detected not only on the stratum corneum (SC) surface but also at the bottom of the upper SC-pH zone, just above the middle SC-pH zone. In inflamed conditions, *S. aureus* proliferates in the thickened upper SC-pH zone but does not penetrate the middle SC-pH zone. When the SC-pH zones are neutralized, *S. aureus* invades all SC-pH zones, reaching the living epidermal layers and causing severe inflammation. Panel adapted from Reference 49 (CC BY 4.0).

Table 2 Beneficial host-microbe interactions by species of skin bacteria

Applied bacterial strain	Origin of isolates	Effects	Experimental model	Reference(s)
<i>Staphylococcus epidermidis</i> NIHLM087	Healthy human skin	Limits pathogen invasion and promotes wound repair by inducing IL-17A ⁺ CD8 ⁺ T cells via dermal dendritic cells	Murine	151–153
		Induces IFN- γ ⁺ CD4 ⁺ T cells via keratinocyte-intrinsic MHC-II expression for tissue immune surveillance	Murine	181
		Induces mucosal-associated invariant T (MAIT) cells by synthesizing riboflavin-derived antigens and promoting wound healing	Murine	182
<i>S. epidermidis</i> JK16	Healthy human nasal cavity	Secretes serine protease Esp to inhibit <i>Staphylococcus aureus</i> biofilm formation and <i>S. aureus</i> colonization	Human	183
<i>S. epidermidis</i> S25	Healthy human skin	Improves skin barrier function by secreting sphingomyelinases that increase skin ceramides	Murine	184
<i>S. epidermidis</i> 1457	Human blood	Induces antimicrobial peptide expression by activating Toll-like receptor 2 signaling	Murine	180
<i>Staphylococcus hominis</i> A9	Healthy human skin	Prevents <i>S. aureus</i> –mediated skin inflammation by killing <i>S. aureus</i> and inhibiting <i>S. aureus</i> toxin production	Human, murine	185, 219
<i>S. hominis</i> C5	Healthy human skin	Prevents <i>S. aureus</i> –induced inflammation by inhibiting <i>S. aureus</i> accessory gene regulator activity regarding quorum sensing	Murine	185, 223
<i>Staphylococcus lugdunensis</i>	Nasal cavity of healthy volunteer	Produces cyclic peptide antibiotics that kill <i>S. aureus</i>	Murine, rat	186
<i>Staphylococcus cohnii</i>	Wild-type murine skin Healthy human skin	Enhances local glucocorticoid synthesis in the skin Ameliorates inflammatory reactions	Murine	81

the expression of CCL20 in HF^s and augments the accumulation of Tregs in neonatal skin; these actions help establish tolerance to bacterial antigens and regulate immune responses (119, 179).

Several skin microorganisms exhibit unique functions and interactions with their hosts (Table 2). Notably, certain strains of *Staphylococcus epidermidis* enhance the production of antimicrobial peptides in the host epidermis through Toll-like receptor 2, activating the host's innate immune defenses against pathogens (180). Th1 cells are accumulated by *S. epidermidis* via keratinocyte-intrinsic MHC-II expression (181). Furthermore, *S. epidermidis* induces nonclassical MHC-Ib–restricted Tc17 (IL-17A–producing CD8⁺ T) cells via dendritic cells in the skin; this

induction helps prevent the invasion of pathogenic bacteria such as *Candida* (153). These Tc17 cells accumulate in the epidermis in response to *S. epidermidis*-derived N-formylated peptides, and tissue injury triggers the production of type 2 cytokines, such as IL-13, in response to tissue alarmins, resulting in the acceleration of tissue repair (151, 152). Additionally, skin Trm17 cells induced by *S. epidermidis* are crucial for accelerating wound healing (157). MAIT (mucosal-associated invariant T) cells, stimulated by riboflavin derivatives from the microbiota, including *S. epidermidis*, also contribute to wound repair (182). Proteases from *S. epidermidis* inhibit biofilm formation and reduce the adhesion of pathogenic bacteria (183), while sphingomyelinase from *S. epidermidis* boosts sphingomyelin metabolism and increases ceramide content in the SC, improving physical barrier function (184).

Staphylococcus lugdunensis and *Staphylococcus hominis*, coagulase-negative staphylococci that colonize the skin, inhibit the growth of *Staphylococcus aureus* by producing antibiotic lugdunin and lantibiotics, respectively (185, 186). The skin, as a steroid-producing organ (187–190), benefits from the anti-inflammatory effects of the commensal skin bacterium *Staphylococcus cohnii*, which enhances local glucocorticoid production (81). *Malassezia*, the predominant fungal species on the skin, metabolizes sebum to release short-chain fatty acids with broad-spectrum antimicrobial properties (191). Certain species of *Malassezia* also drive type 17 immunity in the skin by inducing IL-17 production from $\alpha\beta$ T cells, $\gamma\delta$ T cells, and ILCs (192, 193). The immunosurveillance of these normally harmless fungal colonizations helps inhibit pathogen growth (194, 195). Therefore, various immune responses beneficial to the host are elicited by microorganisms, and skin homeostasis is maintained.

7.3. Inflammatory Skin Diseases Caused by Host-Microbe Interactions

While interactions between host and skin microorganisms generally support homeostasis, the interactions' disruption, termed dysbiosis, can lead to inflammatory skin diseases and infections. Dysbiosis is influenced by both intrinsic factors, such as age and genetic background, and extrinsic factors, such as the use of antimicrobial agents and soaps (196). AD is the most common inflammatory skin disease associated with dysbiosis, characterized by the overgrowth of *S. aureus*; this overgrowth correlates with increased disease severity and decreased microbial diversity (3, 197, 198). The overproliferation of *S. aureus* is known to induce inflammation in various mouse models of AD (81, 199). Toxins produced by *S. aureus*, such as δ -toxin, α -toxin, and PSM α (phenol-soluble modulins α), elicit both innate and adaptive immune responses that contribute to skin inflammation (81, 200–202). *S. aureus*-specific CD4⁺ Trm cells or $\gamma\delta$ Trm cells play a role in the protection against *S. aureus* colonization but also in the exacerbation of AD-like inflammation in mice (203). The interface between the skin microbiome and the neuroimmune axis, particularly as the interface concerns *S. aureus*, is gaining attention. Pore-forming toxins such as PSM (phenol-soluble modulin) produced by *S. aureus* cause pain by directly activating sensory neurons via TRPV1 (transient receptor potential vanilloid 1) in the skin (204). While Th17 cells induced by the topical association of *S. aureus* colocalize with sensory nerve fibers within the dermis under steady-state conditions, commensal-specific Th17 cells release IL-17A upon injury, promoting nerve repair by signaling directly to peripheral sensory neurons via IL-17 receptor A (205). Additionally, proteases produced by *S. aureus* can directly stimulate nerves, causing the characteristic itching associated with AD (206). Notably, *S. aureus* also resides as a commensal bacterium on the skin of more than 30% of healthy individuals, typically at low and barely detectable levels under steady-state conditions (207–209). These findings suggest that under certain environmental conditions, *S. aureus* can act as a catalyst for inflammation. The identification and mitigation of the triggers that enhance the inflammatory actions of *S. aureus* could help in the management of the pathogenesis of various diseases associated with this bacterium.

Thus, understanding the contributing factors and the effects of the microenvironment on the skin microbiota is crucial for unraveling the pathogenesis of microorganisms such as *S. aureus* and *Malassezia* in AD and in other inflammatory skin diseases.

7.4. Understanding the Stratum Corneum Microenvironment for Skin Microbiota

Because the behavior of microorganisms on the skin is profoundly influenced by their microenvironment, an understanding of the conditions under which each bacterium thrives or becomes pathogenic is crucial. The SC provides a specialized microenvironment that shapes the microbial composition. This environment is determined by various factors, including corneocytes, intercellular lipids, secretions from sweat and sebaceous glands, and external stimuli, all of which are further influenced by the host's genetic background, the host's metabolic activity, and any present inflammation. Studies have shown that the skin of patients with congenital ichthyosis and the non-lesional skin of patients with AD exhibit microbial compositions that differ significantly from those of healthy individuals (3, 210). Thus, the microenvironment of the SC not only determines which microbiota can reside there but also regulates the metabolic activities of these microorganisms.

A thorough understanding of the microenvironment of the SC and the microbial habitat it provides is essential for elucidating the functions of the skin barrier. Traditional imaging methods such as SEM have typically shown microorganisms proliferating on the top layer of the SC or just below the desquamating corneocytes, leading to the assumption that the outermost surface of the SC is the primary habitat for microbiota. However, it is hypothesized that commensal skin bacteria might infiltrate deeper into the skin layers, such as the SC, epidermis, dermis, and even adipose tissue, although their exact locations in a living state remain unclear (211).

To precisely identify the localization of specific skin microbiota within the SC-pH zones, a novel method involving dual intravital imaging of SC-pH-labeled and fluorescent protein-labeled bacteria has been developed (49). This technique has allowed the visualization of the behavior of specific bacteria within the SC, revealing that colonization and proliferation by *S. aureus* occur not only on the SC surface but also at the bottom of the neutral upper SC-pH zone, just above the acidic middle SC-pH zone (**Supplemental Video 7**). While the upper SC is more acidic under germ-free conditions, bacterial colonization tends to neutralize it, suggesting that microbial metabolic activity modifies the SC-pH. Additionally, the acidic middle SC-pH zone, termed the acid mantle layer, serves as a critical defense mechanism preventing deeper invasion by *S. aureus* (**Figure 3b**). The compromise of skin barrier function leads to pathogen entry and subsequent microenvironmental changes such as increased water loss, altered sebum production, and induction of inflammation. Such inflammation disrupts the SC-pH zones, neutralizing them and allowing bacteria to penetrate deeper into the living epidermal layers (**Figure 3b**).

Thus, dual intravital imaging of host factors and microorganisms can unveil novel host-microbe interactions, including positional information, and reveal new biological insights into skin diseases. Furthermore, the simultaneous observation of multiple microorganisms will provide important insights into their function as a microbial ecosystem, revealing potential microbe-microbe interactions such as mutualistic relationships or niche and nutrient competition.

7.5. New Treatment Approach Using Live Bacteria: Bacteriotherapy

Recent therapeutic strategies involving bacteria have begun to focus on the intricate interactions between host-microbe and microbe-microbe relationships. These approaches encompass the use of live bacteria, dead bacteria and their components, purified bacterial metabolites and enzymes,

Supplemental Material >

and prebiotics (212, 213). Traditionally, the focus has been on purifying beneficial antibiotics and other microbial compounds for use as pharmaceutical agents. However, an emerging treatment strategy termed bacteriotherapy utilizes live bacteria and their biologically active products, both of which are believed to offer therapeutic benefits.

The primary advantage of using live bacterial products is that these microbes can colonize the host, continuously producing effective metabolites or engaging in beneficial interactions with the host and other members of the skin microbiota. Because these interactions are often strain specific rather than species specific (214), the precise selection of strains for therapeutic use is of critical importance (215). Currently, numerous live bacterial products are in development, with some already undergoing clinical trials. These trials have yielded mixed results regarding the efficacy of oral probiotics and prebiotics in treating skin diseases (216–221).

Notably, treatments with topical *Roseomonas mucosa* (gram-negative rods isolated from the skin of healthy individuals) have been reported to reduce the severity of AD in both children and adults (218, 222). Additionally, allogeneic transplants of *S. hominis* isolated from healthy individuals and autologous transplants of CoNS (coagulase-negative staphylococci)-producing antimicrobial peptides isolated from unaffected skin of patients with AD are being considered for near-future clinical applications (219, 220, 223). Another promising bacteriotherapy involves *Nitrosomonas eutropha*, an ammonia-oxidizing bacterium isolated from soil, which has shown potential in reducing the survival of pathogenic bacteria, including *S. aureus*. Its topical application improved disease severity and reduced pruritus in patients with AD during a phase 2b trial (224).

While these therapies primarily target *S. aureus*, this bacterium is not the sole cause of AD flare-ups in all patients, highlighting the need for methods to select suitable patients for live bacteria treatments. Future developments may include the creation of live bacterial products with broad anti-inflammatory activities and with the capability to directly interact with the host and modulate a range of inflammatory responses. One such potential development involves *S. cobnii*, which has been shown to enhance local glucocorticoid synthesis and exert comprehensive anti-inflammatory effects (81). As these innovative bacteriotherapies continue to evolve, they may significantly affect dermatological practice by harnessing novel host-microbe interactions.

8. CONCLUSION

The skin barrier protects delicate bodies from external harm and prevents the loss of moisture. It consists of four components: the skin microbiota (microbial barrier), the SC (air-liquid barrier), TJs (liquid-liquid barrier), and the LC network (immunological barrier). The interplay between these components maintains homeostasis through an integrated skin barrier function. Furthermore, the skin is not flat but is more reminiscent of a forest because of the presence of hair. Hair defines mammals, and the area surrounding hair follicles provides a unique three-dimensional immune microenvironment. Understanding the sites and skin barrier components wherein immune responses occur is crucial for grasping immune phenomena both in health and in inflammatory skin diseases.

SUMMARY POINTS

1. The epidermis is the outer layer of the skin and consists of four barrier elements: the skin microbiota (microbial barrier), the stratum corneum (SC; air-liquid interface barrier), tight junctions (TJs; liquid-liquid interface barrier), and the Langerhans cell (LC) network (immunological barrier).

2. TJs seal the intercellular spaces at the second layer of the stratum granulosum, forming a liquid-liquid interface barrier that limits molecular movement through the paracellular pathway, and play a vital role in maintaining SC barrier homeostasis.
3. LCs reside beneath TJs in a steady state but extend their dendrites beyond TJs upon activation to capture external antigens at the tips of the dendrites.
4. The SC is composed of intercorneocyte lipid lamellae and corneocytes, both of which are enclosed within a corneocyte lipid envelope and a cornified cell envelope, and keratin filaments. This composition prevents excessive water loss and the entry of harmful substances from the environment.
5. Corneocytes undergo differentiation and develop a three-tiered pH zonation (moderately acidic lower zone, acidic middle zone, and nearly neutral upper zone) that maintains SC thickness and barrier function, contributing to overall SC homeostasis.
6. Hair follicles regulate the recruitment and positioning of immune cells involved in both innate and acquired immunity and facilitate host-microbe symbiosis.
7. The skin microbiota is essential for maintaining skin homeostasis by regulating immune responses and protecting against pathogens.
8. New therapeutic approaches for inflammatory skin diseases using live bacteria that benefit the host are under development for clinical use.

FUTURE ISSUES

1. Identification of the molecular mechanism by which Langerhans cells extend their dendrites beyond the tight junction (TJ) barrier to capture antigens would provide new strategies for regulating percutaneous sensitization to control or prevent allergic disorders.
2. Since epidermal nerve endings remain extended outside the TJ and induce Ca^{2+} signaling in atopic dermatitis, an understanding of the mechanism of homeostatic pruning of these nerve endings is crucial for the development of treatments for pathological itch in inflammatory skin diseases.
3. Elucidation of (a) the molecular switch to initiate corneoptosis, (b) the ion channels responsible for intercellular Ca^{2+} elevation (corneoptosis phase I), and (c) the molecular mechanisms of acidification (corneoptosis phase II) of SG1 cells is crucial to understand corneoptosis, which is the unique cell death process of keratinocytes. This understanding will allow us to develop new strategies to regulate stratum corneum (SC) barrier function.
4. Decipherment of the mechanisms underlying the generation of the three stepwise SC-pH zones and SC-pH regulation will provide novel insights for developing treatments for inflammatory skin diseases and ichthyoses.
5. Intravital imaging and multi-omics analysis will lead to a better understanding of host-bacteria interactions at the species and strain levels and, by extension, of the role of the microbiome in skin homeostasis and inflammatory skin diseases.

6. A deeper comprehension of microenvironments that foster beneficial commensal bacteria and host-microbe interactions will advance the development and efficacy of novel bacteriotherapies.
7. Optimization of the administration of anti-inflammatory microorganisms and substances that promote their colonization would represent a significant therapeutic advancement.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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1. Excellent review of the current understanding of the pathophysiology of atopic dermatitis.

6. First study to demonstrate that filaggrin mutations are significant genetic risk factors for atopic dermatitis.

12. Demonstrated that SG1 cell death requires intracellular Ca^{2+} elevation and subsequent rapid intracellular acidification.

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